

第10章作业

10.2. 即有由Y参数性质. 即有

$$\dot{I}_1 = Y_{11} \dot{U}_1 + Y_{12} \dot{U}_2$$

$$\dot{I}_2 = Y_{21} \dot{U}_1 + Y_{22} \dot{U}_2$$

$$Y_{12} = Y_{21}$$

由(a), 有 $2A = Y_{11} \cdot j \cdot 2V + Y_{12} \cdot (-10j)$ 由(b), $-j1A = Y_{22} \cdot 1V$

$$-j5A = Y_{21} \cdot j \cdot 2V + Y_{22} \cdot (-10j)$$

$$\Rightarrow Y_{11} = -j5S \quad Y_{21} = -\frac{5}{2}S \quad Y_{12} = -j5S \quad \Rightarrow Y_{22} = -jS$$

$$Y_{21} = Y_{12} = -\frac{5}{2}jS$$

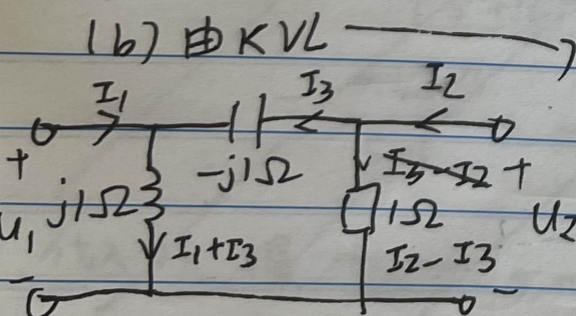
$$Y_{11} = -12.5 + 24j$$

10.3 (a) 由KVL, 有 $\begin{cases} U_1 = I_1 R + (I_1 + I_2) \cdot 2R \\ U_2 + 3U_1 - I_2 R - (I_1 + I_2) \cdot 2R = 0 \end{cases}$

$$\text{为} \begin{pmatrix} -12.5 + 24j & -2.5 + 5j \\ -2.5 + 5j & -j \end{pmatrix}$$

$$\Rightarrow \begin{cases} U_1 = I_1 \cdot 3R + I_2 \cdot 2R \\ U_2 = I_1 \cdot -7R + I_2 \cdot -3R \end{cases}$$

$$\text{Z参数矩阵为} \begin{pmatrix} 3R & 2R \\ -7R & -3R \end{pmatrix}$$



$$\text{有 } U_1 = (I_1 + I_3) \cdot j1\Omega$$

$$U_2 = (I_2 - I_3) \cdot 1\Omega$$

$$I_3 = I_2 - I_1$$

$$I_3 \cdot (-j1\Omega) + (I_1 + I_3) \cdot j1\Omega - (I_2 - I_3) \cdot 1\Omega = 0$$

$$\text{消去 } I_3, \text{ 得 } U_1 = (1+j)I_1 + I_2 j$$

$$U_2 = I_2 - I_2 + I_1 j$$

$$\text{Z参数矩阵为} \begin{pmatrix} 1+j & j \\ j & 1 \end{pmatrix} \Omega$$

0.5 S 断开时, $\therefore U_3 = 9V$ $U_1 = 5V$ $\therefore I_1 = \frac{U_3 - U_1}{R} = +1A$

$U_2 = 3V$ $I_2 = 0V$

有 ~~$U_1 = A_{11} U_2 +$~~

同时代入 $U_1 = A_{11} U_2 + A_{12} (-I_2)$

$I_1 = A_{21} U_2 + A_{22} (-I_2)$

S 接通时, 有

$U_1 = 4V$ $I_1 = \frac{U_3 - U_1}{R} = 1A$

$U_2 = 2V$ $I_2 = -\frac{1}{3}A$

即有 $5 = A_{11} \cdot 3$

$1 = A_{21} \cdot 3$

$4 = A_{11} \cdot 2 + \frac{1}{3} A_{12}$

$1 = A_{21} \cdot 2 + \frac{1}{3} A_{22}$

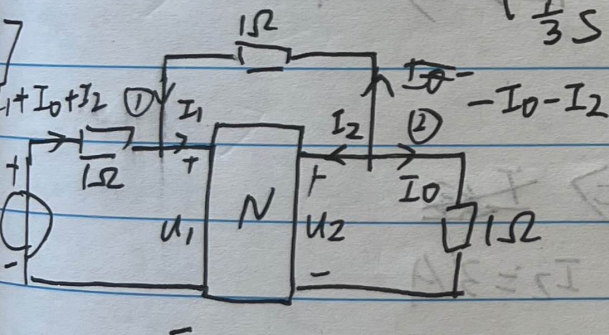
$A_{11} = \frac{5}{3}$

$\Rightarrow A_{21} = \frac{1}{3}$

$A_{12} = 2$

$A_{22} = 1$

即 A 参数矩阵为 $\begin{pmatrix} \frac{5}{3} & \frac{1}{3} \Omega \\ 2S & 1 \end{pmatrix}$



由 KCL, 知各支路电流为如图所示

由 KVL, 知

$24V - (I_1 + I_0 + I_2) \cdot 1\Omega - U_1 = 0$

$U_2 - I_0 \cdot 1\Omega = 0$

$I_1 = 0.4U_1 - 0.2U_2$

$I_2 = -0.2U_1 + 0.6U_2$

$24V - (I_1 + I_0 + I_2) \cdot 1\Omega - (I_0 + I_2) \cdot 1\Omega - I_0 \cdot 1\Omega = 0$

$I_1 = 4A$ $I_2 = 1A$ $U_1 = 13V$ $U_2 = 6V$

$\Rightarrow I_0 = 6A$

10.12 由 KVL, 有 $14V - 2\Omega \cdot I_1 - U_1 = 0$

$$U_2 + I_2 \cdot 1\Omega = 0$$

由端口特性 $U_1 = 4I_1 + 2I_2$

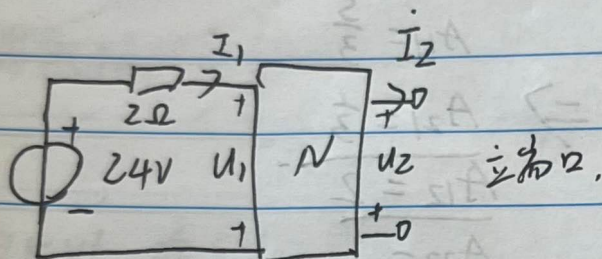
$$U_2 = 4I_1 + 5I_2$$

$$I_1 = 3A \Rightarrow U_1 = 8V$$

$$I_2 = -2A \quad U_2 = 2V$$

10.13

对



开路时有 $U_1 = 6\Omega \cdot I_1$

$$U_2 = 4\Omega I_1$$

$$\Rightarrow U_2 = 12V \Rightarrow U_{开} = 12V$$

$$\text{有 } 24V - 2\Omega \cdot I_1 - U_1 = 0$$

短路时有 $U_1 = 6\Omega \cdot I_1 + (-4) \cdot I_2$

$$U_2 = 4\Omega I_1 + (-6) I_2 = 0$$

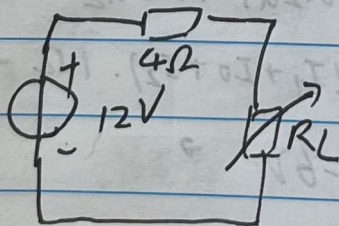
$$\Rightarrow I_{短}$$

$$24V - 2\Omega I_1 - U_1 = 0$$

$$I_2 = 3A$$

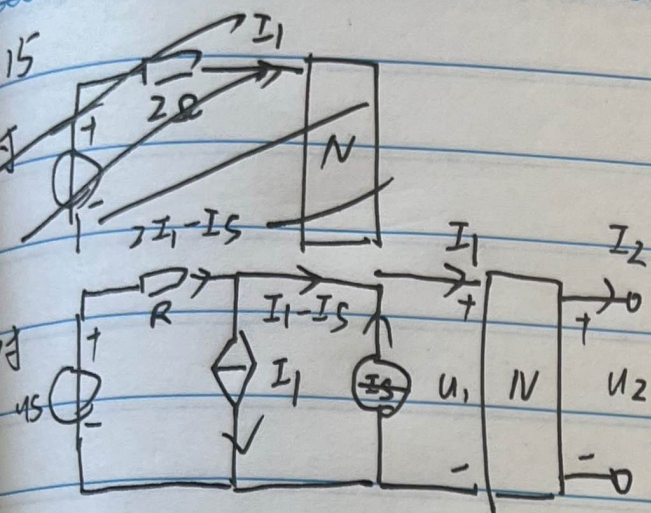
$$\text{等效 } R_{内} = \frac{U_2}{I_2} = 4\Omega$$

等效电路图为



有 $\therefore$ 当 $R_L = 4\Omega$ 时, 最大功率

$$P_{max} = \frac{U^2}{4R_{内}} = 9W$$



开路时, 有 $U_1 = U_2$

$$I_1 = 3S \cdot U_2$$

由 KVL, 有 $U_S - (2I_1 - I_S) \cdot R - U_1 = 0$

$$U_S - (2I_1 - I_S) \cdot R - U_1 = 0$$

$$\text{联立有 } U_2 = \frac{2}{7} V$$

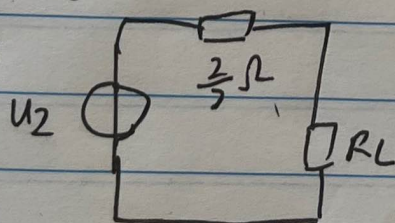
短路时, 有 $U_1 = 2 \cdot I_2$

$$U_S = I_1 = 4 \cdot I_2$$

由 KVL, 有 $U_S - (2I_1 - I_S) \cdot R - U_1 = 0$

$$\Rightarrow I_2 = \frac{1}{5} A \quad \therefore R = \frac{U}{I} = \frac{2}{7} \Omega$$

$\therefore$ 等效电路为



P 最大时

$$R_L = \frac{2}{7} \Omega$$

$$P_m = \frac{U_2^2}{4R_L} = \frac{1}{70} W$$